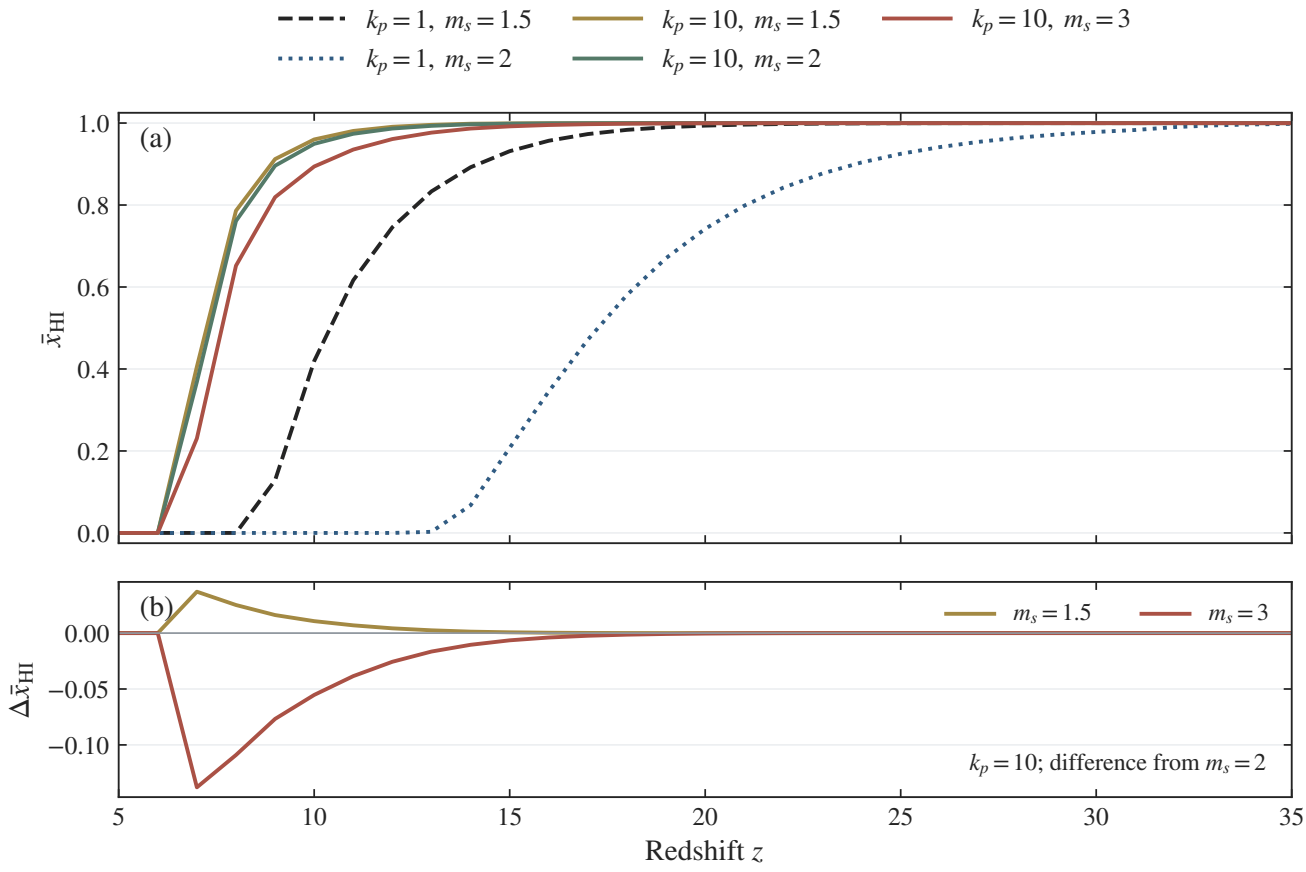
$k_p = 1,\ m_s = 1.5$
$k_p = 10,\ m_s = 1.5$
$k_p = 10,\ m_s = 3$
$k_p = 1,\ m_s = 2$
$k_p = 10,\ m_s = 2$
(a)
1.0
0.8
0.6
0.4
0.2
0.0
$\bar{x}_{\rm HI}$
(b)
$m_s = 1.5$
$m_s = 3$
0.00
−0.05
−0.10
$\Delta\bar{x}_{\rm HI}$
$k_p = 10$; difference from $m_s = 2$
5
10
15
20
25
30
35
Redshift z